E-Commerce Recommendation pipeline

REVIEW – 1

Batch 19 - Section 4

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Abstract

E-commerce platforms provide a very large number of products, which can make it difficult for customers to find products that match their needs and interests. To solve this problem, recommendation systems study different types of user activities, such as searches, clicks, product views, ratings, cart activity, and previous purchases. These activities help the system understand what a customer is interested in. The collected information is then processed through a recommendation pipeline, where the data is collected, cleaned, prepared, and analyzed. Based on this processed data, suitable products are identified, ranked according to their relevance, and recommended to the customer. The recommendations are personalized because they are based on the individual customer's interests and behavior. This project focuses on understanding how an e-commerce recommendation system works, the different stages involved in its data pipeline, and how these personalized recommendations can improve product discovery, customer experience, and make online shopping easier and more convenient.

Introduction

- E-commerce has become a major part of everyday shopping, with platforms offering thousands or even millions of products. With so many choices available, customers may find it difficult to identify products that are relevant to their needs.

- An e-commerce recommendation system helps solve this problem by understanding customer interests and suggesting products that they are likely to prefer. It uses information such as search history, product views, clicks, ratings, cart activity, and previous purchases.

- Behind these recommendations is a recommendation pipeline. It takes user and product data, processes the information, finds relevant products, ranks them, and finally displays the most suitable recommendations to the customer.

- Therefore, recommendation systems help make online shopping more personalized, convenient, and efficient, while also helping e-commerce platforms improve customer engagement and product discovery.